

ARNAV CHUDIWALE

Supply Chain & Operations | Network Optimization | Data Analytics | Public-Sector AI

Stillwater, OK | +1 (703) 203-6481 | arnav.chudiwale@okstate.edu

linkedin.com/in/arnav-chudiwale | portfolio-pi-indol-72.vercel.app

Open to relocation: UAE, Saudi Arabia, Europe, Singapore | Available June 2027

SUMMARY

Industrial engineer who turns operational decisions into models leaders can approve. At **EY**, worked on site with the Government of Maharashtra on a **\$600M World Bank-funded program**, testing, redesigning and costing a generative-AI advisory chatbot for farmers. Builds network optimization, routing, scheduling and business-case models in Python and Gurobi, and states every recommendation with its trade-offs and risks.

EDUCATION

Oklahoma State University, Stillwater, OK
M.S., Industrial Engineering & Management

Jan 2025 – May 2027 (expected)

Coursework: Linear Optimization; Network Optimization; Operations Research; Stochastic Processes; Discrete-Event Simulation; Supply Chain Strategy; Advanced Data Analytics for Industrial Engineering; Fundamentals of GIS

Thapar Institute of Engineering & Technology, Patiala, India
B.E., Electrical & Computer Engineering

Nov 2020 – Nov 2024

PROFESSIONAL EXPERIENCE

Technology Consultant Intern | Ernst & Young (EY) LLP, Government Advisory

Jan 2024 – Jun 2024

On site at the Government of Maharashtra, Department of Agriculture, Mumbai | Project on Climate Resilient Agriculture (PoCRA), a \$600M World Bank-funded program

- Worked inside the client's office as part of EY's project-management team, reporting to the program's **Project Director, a senior civil servant (IAS)**, on the IT strategy for the program's next phase.
- **Generative-AI advisory chatbot:** built and tested the first prototype on **Microsoft Azure** (Azure OpenAI, Cognitive Search) with Microsoft's developers; found three reasons it would fail at scale: keyword retrieval that missed the context of farmers' questions, no integration with the program's existing app, and repeated deployment errors.
- Initiated a collaboration with a **Microsoft technology partner** to rebuild the chatbot on **retrieval-augmented generation (RAG)**, designed to follow context and tailor answers by crop and location; the program took this design forward and has built further versions on it since.
- Built the bottom-up **cost model** (users, queries, tokens, compute, database, storage, RAG service): **INR 147,000 (about US\$1,800) a month for 10,000 farmers**, or about INR 15 per farmer.
- Owned the **emerging-technology section** (AI, IoT, drones) of the program's Phase 2 IT strategy, a 3–4 year roadmap reviewed by the Project Director, mapping current processes against target-state solutions.
- Designed two **AI anti-fraud proofs of concept** for subsidy approvals: facial verification for field inspections signed off by stand-ins, and OCR of tax (GST) numbers on vendor bills to expose forced purchasing; both were **formally approved** and handed to the program's IT team for prototyping.
- Opened technical partnership discussions with **IIT Bombay's IoT innovation hub (TiH Foundation)** to scope water- and soil-health monitoring for the next phase.

OPTIMIZATION & SUPPLY CHAIN PROJECTS

Circular Supply Chain Network Design

Jan 2026

Electronics returns and refurbishment network, California | Python, Gurobi, ArcGIS Pro, Power BI

- Designed a returns network for about **28,000 smartphones a year from 54 retail stores**; compared ArcGIS distance-minimizing location-allocation with Gurobi cost-minimizing facility models for one to five sites.
- Found a **two-site cost optimum (Fresno and Reno)** and showed why the closest sites were not the cheapest: fixed facility costs were about two orders of magnitude larger than transport costs.
- Modeled a **\$2.6M three-year NPV** (base case) and **14-month payback** on \$2.25M of capital, net of \$1.55M annual operating cost; grade-based refurbishment raised value recovered per phone from **\$22 to \$141**.
- Stress-tested seven parameters ($\pm 20\%$ prices, volumes, costs and capital): NPV stayed positive in every scenario (**\$0.9M to \$4.1M**), and resale values could fall 25% before it reached zero; built a Python pipeline and Power BI dashboards to present them.

Last-Mile Delivery & Peak-Season Capacity

Dec 2025

City fulfillment center, peak-day routing and dispatch | Python, Google OR-Tools, SciPy

- Built a vehicle-routing model for **500 delivery stops and 32 vehicles**, then tested a **162% demand surge** (1,490 to 3,906 packages a day) to size peak-season capacity.
- Compared **six overtime and outsourcing strategies** against a 2,306-package daily shortfall; a hybrid allocation cut **cost per package by 22%** while serving all demand.
- Designed a **four-wave dispatch strategy** that reuses returning vans; in simulation it served all 3,906 packages with **25% less outsourcing** (1,922 to 1,447) and **5–19% lower peak-day cost** than the best static plan, with no new vehicles.

TEACHING & LEADERSHIP

Instructor, Engineering Programming | Oklahoma State University

Aug 2026 – Present

School of Industrial Engineering & Management

- Teach **Python for data science and advanced Excel** to **27 undergraduate engineers**, turning technical methods into practical problem-solving.

Treasurer | INFORMS Student Chapter, Oklahoma State University

Current

INFORMS is the leading international association for operations research and analytics professionals.

Founder & General Secretary | PLEX, The Product Managers' Club

Aug 2023 – Dec 2023

Thapar Institute of Engineering & Technology

- Founded a student product club and led a cross-department team building a **campus recruitment portal for about 10,000 students**, with the Computer Science department and the placement office.
- Coordinated a smart-classroom automation initiative with the Department of Energy & Environment, targeting **INR 4.4M in annual electricity savings** for one lecture building.

HONORS

Runner-up, National Product Case Competition, BITS Pilani

Mar 2023

Built an M&A strategy for Zuora to counter fraud in the subscription industry; one of 10 national finalists.

Winner, Standard Chartered Bank Ideathon, Thapar Institute

Feb 2023

Pitched UdhaarLo, a small-business credit platform pairing faster loan approval with real-time credit-score updates.

SKILLS

Methods: network optimization, mixed-integer programming, facility location, vehicle routing, scheduling, discrete-event simulation, cost-benefit and DCF modeling, sensitivity analysis, IT and technology strategy

Optimization & simulation: Gurobi, Google OR-Tools, PuLP, NetworkX, Simio, SimPy

Data & analytics: Python (Pandas, NumPy, SciPy, scikit-learn), SQL, R, Excel (Solver, VBA, Power Query), SAS Viya

Visualization & GIS: Power BI, Tableau, JMP, Matplotlib, Plotly, ArcGIS Pro (Network Analyst)

Cloud & AI: Microsoft Azure, Azure OpenAI, retrieval-augmented generation (RAG)

Languages: English (fluent), Hindi (native)